

PocketSmart AI

File 4 — Project Planning

Project: GenAI-Powered Budget & Recommendation Assistant

This document defines the planned development approach for PocketSmart AI, including project phases, work packages, technology planning, testing preparation, and expected deliverables.

1. Project Objectives

- Develop a GenAI-powered assistant that understands a user's budget and requirements.
- Provide relevant product or service recommendations across suitable categories.
- Use structured user inputs such as category, budget, preferences, and requirements.
- Present recommendations in a clear and user-friendly interface.
- Validate the application through functional, usability, AI-response, and integration testing.

2. Development Phases

Phase	Planned Work	Expected Output
1. Brainstorming & Ideation	Define problem, target users, use cases, and core AI concepts.	Project concept and scope
2. Requirement Analysis	Identify functional and non-functional requirements.	Requirement specification
3. Project Design	Design system architecture, UI flow, data flow, and AI integration.	Design specification
4. Project Planning	Prepare timeline, tasks, resources, milestones, and risk controls.	Project plan
5. Project Development	Implement frontend, backend, AI integration, recommendation logic, and data handling.	Working prototype
6. Project Testing	Test features, inputs, AI responses, errors, and overall usability.	Test results and fixes
7. Project Documentation	Prepare technical documentation, screenshots, methodology, and results.	Project documentation
8. Project Demonstration	Prepare presentation, walkthrough, and demonstration video.	Demo video and presentation

3. Work Breakdown Plan

Requirements: Define users, recommendation categories, budget inputs, preferences, expected outputs, and system constraints.

UI/UX: Design the input screens, recommendation display, navigation, and responsive layout.

AI Layer: Connect the application to the selected Gemini model and design prompts for recommendation generation.

Recommendation Engine: Process budget and preference information and format recommendations consistently.

Application Backend: Implement APIs, validation, business logic, and integration between frontend and AI services.

Database / Data Handling: Store required application data and maintain a structured representation of user inputs and results where applicable.

Testing: Prepare test cases for valid inputs, invalid inputs, edge cases, AI output quality, and UI behavior.

Deployment: Prepare production configuration and deploy the completed application.

4. Suggested Milestones

Milestone	Completion Target
M1 — Scope Frozen	Project problem, objectives, and core features finalized
M2 — Requirements Complete	Functional and non-functional requirements documented
M3 — Design Complete	Architecture and interface flow prepared
M4 — Core Prototype	Main user flow and AI recommendation flow working
M5 — Feature Complete	Planned application features integrated
M6 — Testing Complete	Major defects resolved and test results documented
M7 — Documentation Complete	Final report and supporting material prepared
M8 — Demonstration Ready	Presentation and project video completed

5. Technology Planning

Area	Planned Technology / Approach
Frontend	React / modern web UI
Backend	Application API and server-side business logic
Generative AI	Google Gemini API
Data / Backend Services	Cloud database or backend service as required by the implementation
Deployment	Cloud hosting platform suitable for the frontend and backend
Version Control	Git and GitHub
Development	VS Code and local development environment

6. Resource Planning

Human resources: Project developer/team responsible for requirements, UI, coding, AI integration, testing, documentation, and demonstration.

Software resources: Code editor, Python/Node.js or selected development runtime, Git/GitHub, browser, AI development tools, and deployment platform.

AI resources: Gemini API access and carefully designed prompts for recommendation generation.

Testing resources: Representative user scenarios, different budget ranges, categories, preferences, invalid inputs, and edge cases.

7. Risk Management Plan

Risk	Potential Impact	Planned Control
AI API quota / availability	Recommendations may fail or become temporarily unavailable.	Use appropriate model configuration, error handling, retry logic, and fallbacks.
Unexpected AI output	Recommendations may be incomplete or poorly structured.	Use structured prompts, validation, and output formatting.
Invalid user input	Incorrect recommendations or application errors.	Input validation and clear user guidance.
Deployment configuration issue	Application may fail after deployment.	Test production configuration and environment variables before release.
Limited test coverage	Undetected functional issues.	Create test cases covering normal, invalid, and edge-case scenarios.

8. Final Planned Deliverables

- Functional PocketSmart AI application.
- Source code and project repository.
- Project design and planning documentation.
- Test cases and test results.
- Screenshots of the completed application.
- Project presentation and demonstration video.
- Final project report.

Planning note: The exact schedule can be adjusted according to the project's academic submission date, team size, feature scope, and availability of external AI/cloud services.